

Subject Description Form

Subject Code	AF3625
Subject Title	Engineering Economics
Credit Value	3
Level	3
Exclusion	AF2618
Objectives	This subject aims to equip students with 1. The fundamental concepts of micro- and macroeconomics related to the engineering industry; 2. The fundamental understanding of finance and costing for engineering operations, budgetary planning and control.
Intended Subject Learning Outcomes	Upon successful completion of this subject, students will be able to: 1. Understand how the relevant economic factors shape the environment within which an engineering company operates; 2. Evaluate the financial condition of a company; 3. Apply the basic cost accounting techniques in the planning and control of engineering and production activities.
Subject Synopsis/ Indicative Syllabus	<u>Economic Environment of a Firm</u> Microeconomic Factors Scarcity, choice and opportunity cost; Demand, supply and price; Profit-maximizing behavior of the firm; Organization of industry: perfect competition and monopoly Macroeconomic Factors International trade and globalization <u>Engineering Economics</u> Return on investment; Accounting profit versus economic profit <u>Fundamentals of Budgetary Planning and Control</u> Principle types of budgets for production and service operations; Approaches to budgeting and the budgeting process; Investment and source of finance; Cost of capital; Evaluation of investment alternatives
Teaching/ Learning Methodology	The two-hour lecture each week focuses on the introduction and explanation of key concepts of Engineering Economics. The one-hour tutorial provides students with directed studies to enhance their self-learning capacities. Individual and group activities including discussions and presentations are conducted to facilitate students' understanding and application of the concepts they have learned to tackling real-life problems in Engineering Economics.

Assessment Methods in Alignment with Intended Learning Outcomes	Specific Assessment Methods/Tasks	% Weighting	Intended Subject Learning Outcomes to be Assessed (Please tick as appropriate)		
			1	2	3
	Continuous Assessment	50%			
	1. In-class activities	15%	√	√	√
	2. Written assignments	15%	√	√	√
	3. Test	20%	√	√	√
	Final Examination	50%	√	√	√
	Total	100 %			
Student Study Effort Expected	Class contact:				
	• Lecture				26 Hours
	• Tutorial				13 Hours
	Other student study effort:				
	• Study and self-learning				48 Hours
	• Presentation preparation and written assignments				18 Hours
	Total student study effort:				105 Hours
Reading List and References	Recommended Textbooks 1. Parkin and Bade, <i>Foundations of Microeconomics</i> , 8 th ed., Pearson, 2018. 2. Sullivan, Wicks and Koelling, <i>Engineering Economy</i> , 17 th ed., Pearson, 2019. References 1. Robert H. Frank, <i>The Economic Naturalist: Why Economics Explains Almost Everything?</i> , Basic Books, 2011.				
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